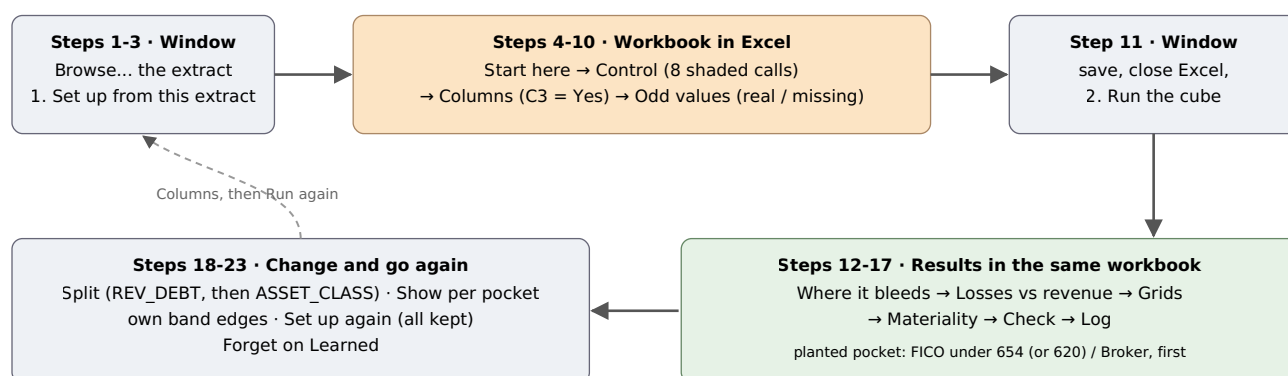


Origination analysis: the window and the workbook

Who this is for: the analyst preparing an origination analysis for a bank's line of business (LOB). No command line is needed. **What you finish with:** one workbook beside the loan extract. It holds your answers, a ranked list of the pockets of the book that lose more than their share, losses set against revenue, and, if you ask for it, every pocket split by a third column. **Walked:** 25 September 2026, on commit 00c0bdd, with a synthetic book of 8,000 loans. The book hides two things: a bad pocket (scores under 620 in the Broker channel), and borrowers carrying more revolving debt than usual for their score, who go bad 1.8 times as often. No real bank data was used.

What changed since the second walk: the Split column on Columns, the Losses vs revenue and Materiality tabs, Show per pocket, and band edges typed with semicolons. Steps 13, 15 and 18 to 20 are new.



Peach = where you answer (Excel). Grey = the window. Green = where you read the results. Nothing is typed at a command line.

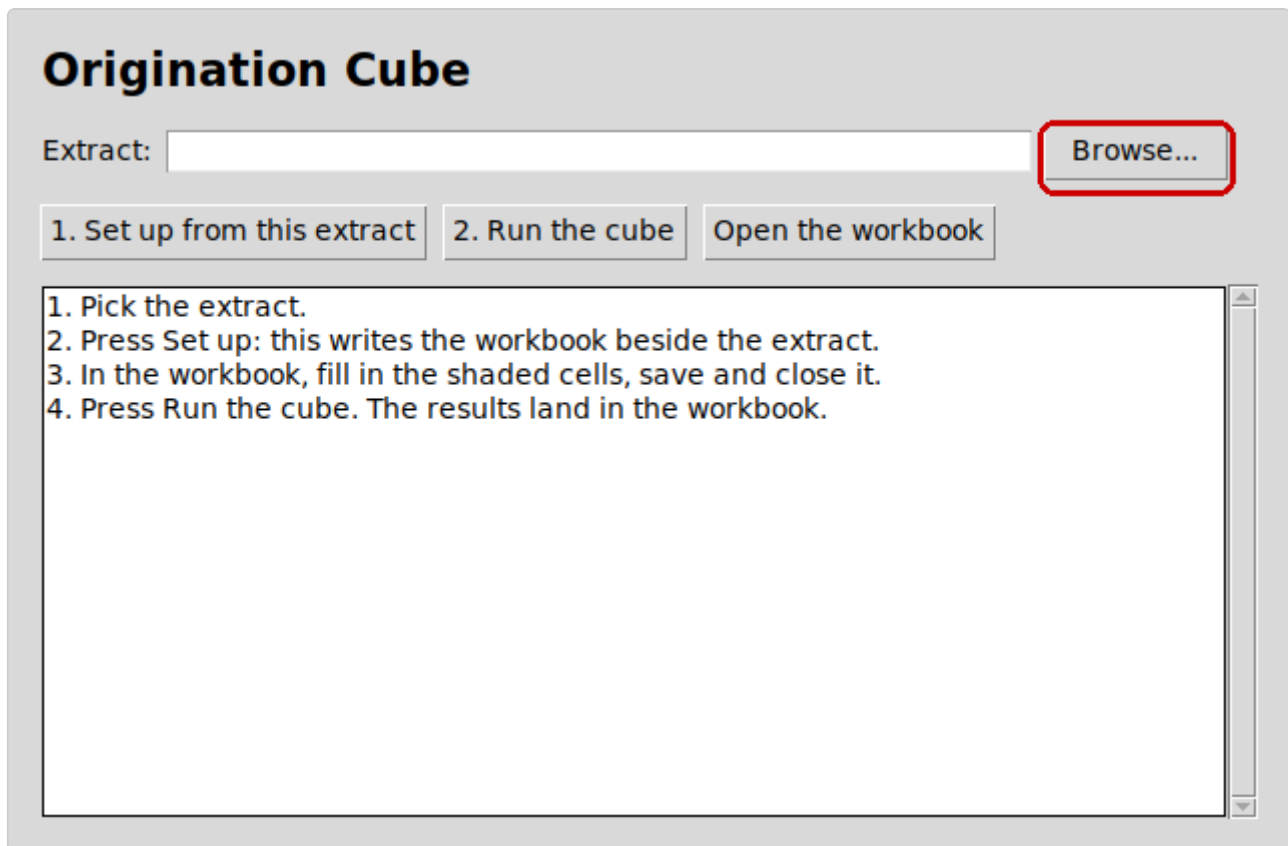
The route: two buttons in the window, one workbook in Excel.

Before you start

- **Python 3.10 or later**, and **Install add-ons.bat** run once. You only need to do this the first time.
- **Excel.**
- **The loan extract** (.csv or .xlsx) in its own folder. It needs a loan number, the booked amount, a yes/no outcome, GCO dollars and RANR dollars.
- **Pick the extract, never the workbook.** After Set up, the folder holds both. The file picker shows both. Picking the workbook sets up a second workbook from the first one (wrong turn K).

Step 1: Open the window

Double-click **Origination Cube.pyw** in the origination-cube folder.



Right: a window titled Origination Cube, an empty Extract box, three buttons, and four numbered lines in the box underneath.

Wrong: if the box says something needs installing, run `Install add-ons.bat` once, then open the window again.

Step 2: Pick the extract

Press **Browse...** and pick the loan file.

Origination Cube

Extract: Browse...

1. Set up from Directory: 

1. Pick the ex

2. Press Set u

3. In the work

4. Press Run

.pytest_cache

configs

docs

src

tests

tools

File name:

Files of type:

On Windows this is the usual Open dialog. The picture shows the same step on the test machine.

Origination Cube

Extract: Browse...

1. Set up from this extract 2. Run the cube Open the workbook

1. Pick the extract.

2. Press Set up: this writes the workbook beside the extract.

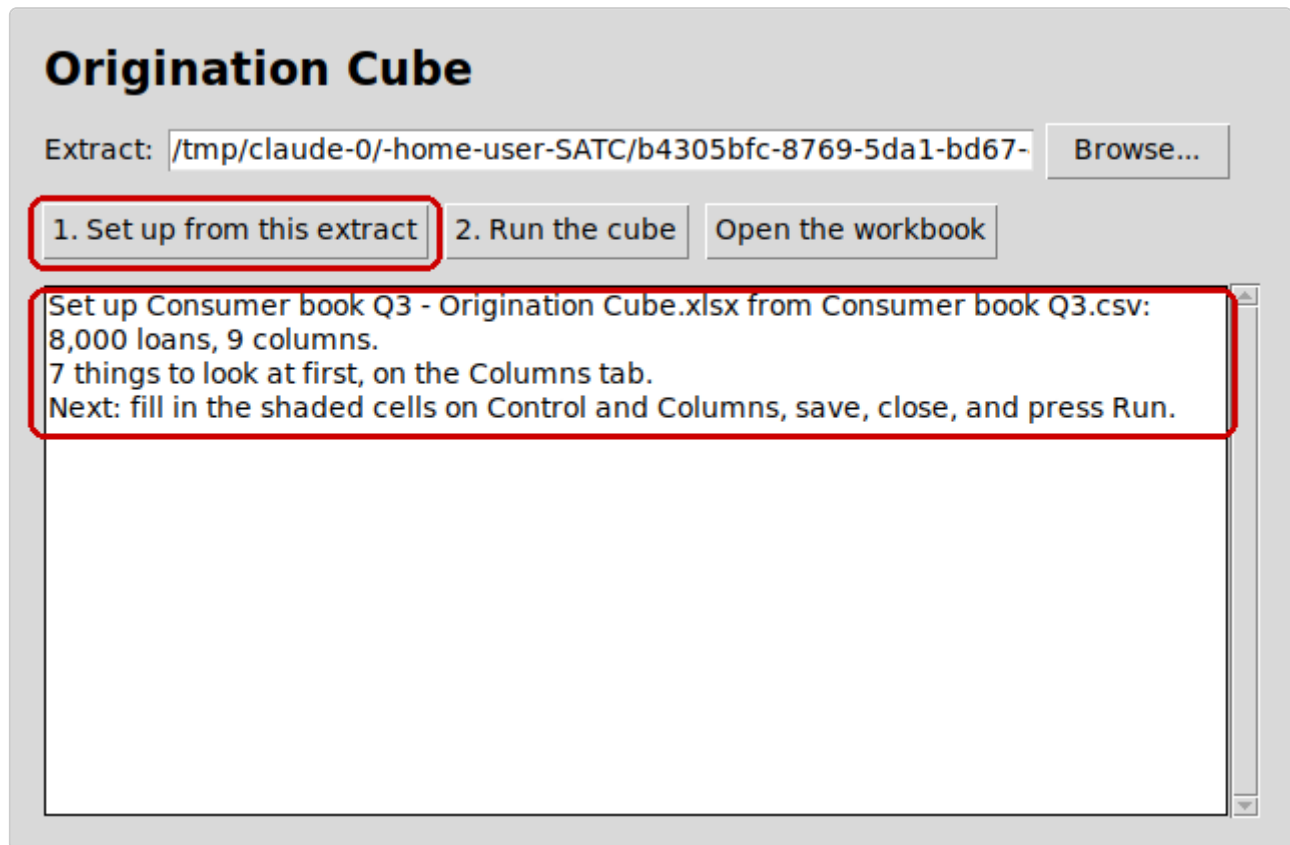
3. In the workbook, fill in the shaded cells, save and close it.

4. Press Run the cube. The results land in the workbook.

Right: the path of the extract is in the Extract box. The box shows the start of the path, so the file name itself may be out of sight. Step 3 names it.

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Step 3: Press 1. Set up from this extract

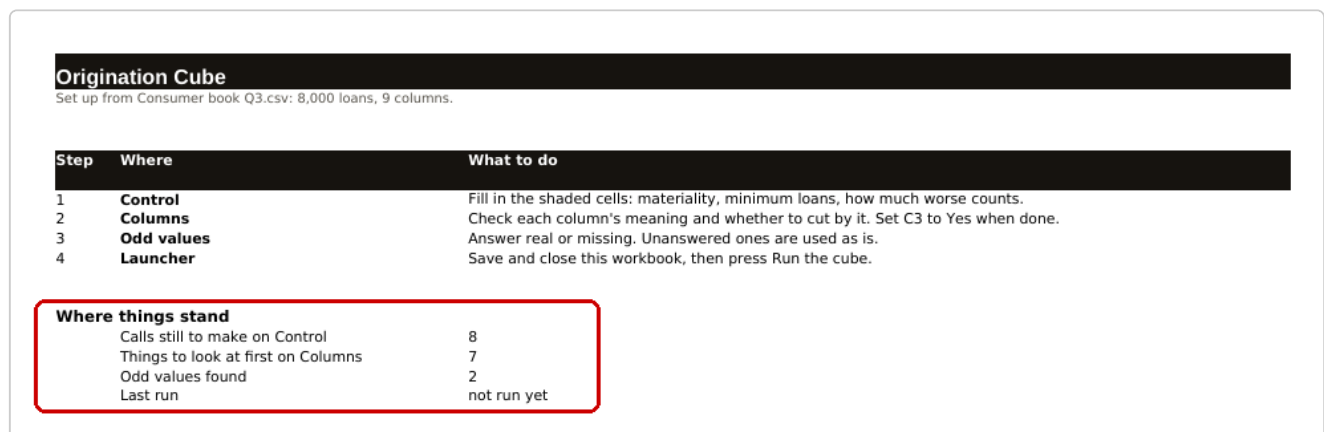


Right: "Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv: 8,000 loans, 9 columns", then "7 things to look at first, on the Columns tab". The workbook now sits beside the extract.

Check the loan count against what the bank sent. If it's wrong, you picked the wrong file.

Step 4: Open the workbook: Start here

Press **Open the workbook**, or double-click it in its folder. It opens on **Start here**.



Right: four steps (Control, Columns, Odd values, Launcher) and Where things stand: 8 calls to make on Control, 7 things to look at on Columns, 2 odd values, not run yet.

Step 5: Control: see what needs an answer

Click the **Control** tab.

Control center

Fill in the shaded cells. The rest have starting values; change them if the population calls for it. Pick from the list, or type a number in the next column to override it. Changes apply on the next Run.

Shaded = still needs an answer

Setting	Change	Or enter your own	In use	What it means
What the cube looks at				
How old a loan must be to count				Needs an answer before we run. Pick one, or enter your own.
How columns are recognised				
A number column with this many values or fewer is a category	12 values (recommended)		12 values	Term (36, 48, 60) or a grade from 1 to 10 is read as categories, not cut into bands.
A text column with more values than this is too fine to cut by	50 values (recommended)		50 values	State (50 or so) is still a dimension; a city or a dealer name is raised as a question instead.
How many bands a number column is cut into	5 bands (recommended)		5 bands	Fine enough to see a trend, with each band holding a fifth of the loans.
Where the band edges fall	Each band holds about the same number of loans (recommended)	n/a	Each band holds about the same number of loans	Every band can be tested equally well. Edges land on odd numbers like 663.
Is it enough loans to matter				
Fewest loans in a pocket before it is tested				Needs an answer before we run. Pick one, or enter your own.
Fewest loans with a loss before a loss rate is tested				Needs an answer before we run. Pick one, or enter your own.
Is it material				
Smallest excess loss worth reporting				Needs an answer before we run. Pick one, or enter your own.
Does it underperform				
What a pocket is judged against		n/a		Needs an answer before we run. Pick one from the list.
How much worse than its comparison a pocket must be				Needs an answer before we run. Pick one, or enter your own.
How much better than its comparison a pocket must be				Needs an answer before we run. Pick one, or enter your own.
Is it real				
How sure a difference must be before it counts				Needs an answer before we run. Pick one, or enter your own.
How often a real gap should be caught	80% of the time (recommended)		80% of the time	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg) (recommended)	n/a	Hold down the share of lucky finds (Benjamini-Hochberg)	Of the pockets that pass, about 5% at 95% are expected to be luck. Suits scanning for leads.

Right: eight rows are shaded peach. They are the calls the tool won't make for you: loan age, fewest loans, fewest losses, materiality, what a pocket is judged against, how much worse, how much better, and how sure. The other rows start on a recommended setting. Grey n/a cells mean that row only takes a pick from the list.

Step 6: Columns: see what each column was taken to be

Click the **Columns** tab.

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done, confirmed meanings carry over to the next extract.

Checked every column?

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (use ; between)	Show per pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)						name contains 'loan'; a different value on every row	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes					name contains 'fico'; 98% of values between 300 and 850	0%	-9999, 713, 716
CHANNEL	Category	Yes					3 different values	0%	Online, Broker, Branch
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes					name contains 'bal'; positive on every loan name contains 'bad'; 0 and 1 (7.1% are 1)	0%	7655.75, 27999.47, 39684.43 0, 2, 1
GCO_AMT	GCO dollars						name contains 'gco'; zero on 93% of loans, never negative	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars						name contains 'ranr'; numbers	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes					4 different values	0%	1, 2, 4
REV_DEBT	Amount or number	Yes					numbers with 7,889 values	0%	15868.86, 12884.7, 8920.97

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done, confirmed meanings carry over to the next extract.

Checked every column?

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (use ; between)	Show per pocket	Split pockets by it?	Look first
LOAN_NBR	Loan number (key)					
FICO	FICO score	Yes				160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.
CHANNEL	Category	Yes				suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes				1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract. 1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.
GCO_AMT	GCO dollars					name contains 'gco'; zero on 93% of loans, never negative
RANR_AMT	RANR dollars					name contains 'ranr'; numbers
ASSET_CLASS	Category	Yes				4 different values
REV_DEBT	Amount or number	Yes				numbers with 7,889 values

Right: one row per column, each with a meaning in What it is and the reason beside it. Seven rows are shaded under **Look first**:

- **FICO:** 160 rows of -9999
- **CHANNEL, ASSET_CLASS** and **REV_DEBT:** suggested from their shape alone
- **BAD_FLAG:** one value that is neither 0 nor 1
- **GCO_AMT:** one value that isn't a number
- **RANR_AMT:** 2,612 negative values

Checked every column? (C3) is shaded and empty. The ringed columns, **Show per pocket** and **Split pockets by it?**, are for steps 18 to 20. Leave them blank for now.

Step 7: Odd values, and Learned

Click **Odd values**.

Odd values

Values that might be codes rather than real numbers. Answer real or missing (missing is treated as blank and counted). Unanswered ones are used as is.

Column	What looks odd	Rows	Answer	Note
FICO	the value -9999 repeated far more than any other	160		
RANR_AMT	negative values in a column that is mostly positive	2,612		

Right: FICO's -9999 (160 rows) and RANR's negatives (2,612 rows), each with a shaded Answer cell.

The **Learned** tab lists what the tool remembers from earlier runs. On the first extract it says Nothing learned yet.

Learned

Meanings and answers carried over from past runs. Set a row to Forget to drop it on the next Run. To stop it being learned again, also change it on Columns.

Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
			Nothing learned yet.			

Step 8: If you press Run too early

If you save, close and press **2. Run the cube** before answering, nothing runs.

Origination Cube

Extract: /tmp/claude-0/-home-user-SATC/b4305bfc-8769-5da1-bd67-

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Control!C6: "How old a loan must be to count" needs an answer. Pick one, or enter your own in column D.
- Control!C13: "Fewest loans in a pocket before it is tested" needs an answer. Pick one, or enter your own in column D.
- Control!C14: "Fewest loans with a loss before a loss rate is tested" needs an answer. Pick one, or enter your own in column D.
- Control!C16: "Smallest excess loss worth reporting" needs an answer. Pick one, or enter your own in column D.
- Control!C18: "What a pocket is judged against" needs an answer. Pick one from the list.
- Control!C19: "How much worse than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
- Control!C20: "How much better than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
- Control!C22: "How sure a difference must be before it counts" needs an answer.

Right: "Couldn't run yet", then one line per thing to fix, each naming the tab and cell. The same list goes to the **Log** tab.

Scroll the box. The last line, Columns!C3: set Checked every column to Yes, is below the fold:

Origination Cube

Extract:

Control!C13: "Fewest loans in a pocket before it is tested" needs an answer. Pick one, or enter your own in column D.

- Control!C14: "Fewest loans with a loss before a loss rate is tested" needs an answer. Pick one, or enter your own in column D.
- Control!C16: "Smallest excess loss worth reporting" needs an answer. Pick one, or enter your own in column D.
- Control!C18: "What a pocket is judged against" needs an answer. Pick one from the list.
- Control!C19: "How much worse than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
- Control!C20: "How much better than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
- Control!C22: "How sure a difference must be before it counts" needs an answer. Pick one, or enter your own in column D.

- Columns!C3: set Checked every column to Yes once you've checked each column's meaning.

Step 9: Fill in the shaded cells on Control

Click each shaded **Choose** cell and pick from its list. To use a number that isn't offered, type it in **Or enter your own** instead. The walk used:

Setting	Answer
How old a loan must be to count	Every loan
Fewest loans in a pocket before it is tested	30 loans
Fewest loans with a loss before a loss rate is tested	10 losses
Smallest excess loss worth reporting	1% of the book's total losses
What a pocket is judged against	The rest of its band
How much worse	1.25 times
How much better	0.8 times
How sure	95% sure

Control center				
Fill in the shaded cells. The rest have starting values; change them if the population calls for it. Pick from the list, or type a number in the next column to override it. Changes apply on the next Run.				
Shaded = still needs an answer				
Setting	Change	Or enter your own	In use	What it means
What the cube looks at				
How old a loan must be to count	Every loan		Every loan	New loans have not had time to go bad, so a pocket full of recent loans looks better than it is.
How columns are recognised				
A number column with this many values or fewer is a category	12 values (recommended)		12 values	Term (36, 48, 60) or a grade from 1 to 10 is read as categories, not cut into bands.
A text column with more values than this is too fine to cut by	50 values (recommended)		50 values	State (50 or so) is still a dimension; a city or a dealer name is raised as a question instead.
How many bands a number column is cut into	5 bands (recommended)		5 bands	Fine enough to see a trend, with each band holding a fifth of the loans.
Where the band edges fall	Each band holds about the same number of loans (recommended)	n/a	Each band holds about the same number of loans	Every band can be tested equally well. Edges land on odd numbers like 663.
Is it enough loans to matter				
Fewest loans in a pocket before it is tested	30 loans		30 loans	The old workbook's value. Stops a test on a handful of loans; the test itself decides whether a gap is luck.
Fewest loans with a loss before a loss rate is tested	10 losses		10 losses	A loss rate rests on how many loans lost money, not how many a pocket holds. Ten is a common floor.
Is it material				
Smallest excess loss worth reporting	1% of the book's total losses		1% of the book's total losses	Scales with the book, so the same setting works on any population.
Does it underperform				
What a pocket is judged against	The rest of its band	n/a	The rest of its band	Broker under 620 against the rest of under 620. Both comparisons are printed; this one decides the flag.
How much worse than its comparison a pocket must be	1.25 times		1.25 times	The old workbook's value. Moderate gaps; more pockets flagged.
How much better than its comparison a pocket must be	0.8 times		0.8 times	The old workbook's value.
Is it real				
How sure a difference must be before it counts	95% sure		95% sure	The usual standard. About 1 pocket in 20 with no real difference will still pass.
How often a real gap should be caught	80% of the time (recommended)		80% of the time	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg) (recommended)	n/a	Hold down the share of lucky finds (Benjamini-Hochberg)	Of the pockets that pass, about 5% at 95% are expected to be luck. Suits scanning for leads.

Right: the peach shading is gone, and In use and What it means fill in for each row.

Watch out: a dollar amount typed in Or enter your own for materiality applies the same dollars to the RANR shortfall as to losses (wrong turn J). Use the percentage options unless you mean that.

Step 10: Confirm the columns and answer the odd values

On **Columns**, check each What it is. Fix any that's wrong from its list, then set **Checked** every column? (C3) to **Yes**.

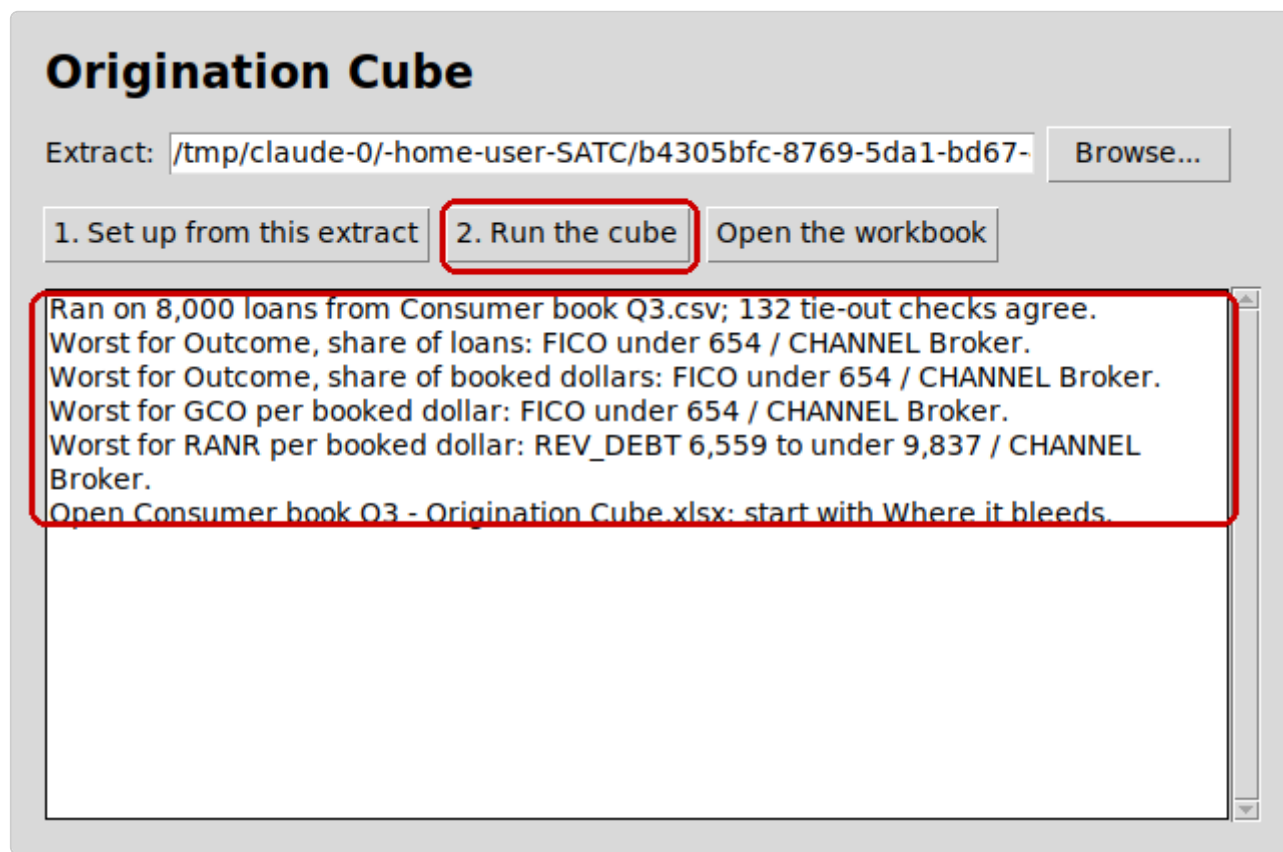
Columns									
Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.									
Checked every column? Yes									
Column	What it is	Cut by it? Yes means (outcome only)	Band edges (use ; between)	Show per pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)						name contains 'loan'; a different value on every row	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes				160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.	name contains 'fico'; 98% of values between 300 and 850	0%	-9999, 713, 716
CHANNEL	Category	Yes				suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.	3 different values	0%	Online, Broker, Branch
ORIG_BAL	Booked amount	Yes				1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	name contains 'bal'; positive on every loan	0%	7655.75, 27999.47, 39684.43
BAD_FLAG	Outcome (yes/no)					1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	name contains 'bad'; 0 and 1 (7.1% are 1)	0%	0, 2, 1
GCO_AMT	GCO dollars					2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.	name contains 'gco'; zero on 95% of loans, never negative	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars					2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.	name contains 'ranr'; numbers	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes				suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.	4 different values	0%	1, 2, 4
REV_DEBT	Amount or number	Yes				suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.	numbers with 7,889 values	0%	15868.86, 12884.7, 8920.97

On **Odd values**, answer each row. The walk said FICO's -9999 is **missing** and RANR's negatives are **real**, because RANR is revenue and can be negative.

Odd values				
Values that might be codes rather than real numbers. Answer real or missing (missing is treated as blank and counted). Unanswered ones are used as is.				
Column	What looks odd	Rows	Answer	Note
FICO	the value -9999 repeated far more than any other	160	missing	
RANR_AMT	negative values in a column that is mostly positive	2,612	real	

Step 11: Save, close, and press 2. Run the cube

Save the workbook and **close Excel**. Then press **2. Run the cube**.



Origination Cube

Extract: /tmp/claude-0/-home-user-SATC/b4305bfc-8769-5da1-bd67- Browse...

1. Set up from this extract 2. Run the cube Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Right:

- "Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree."
- one Worst for line per measure. For the three loss measures it's **FICO under 654 / CHANNEL Broker**, the planted pocket seen through five equal bands.

Check the loan count. It should match step 3.

The Worst for RANR line on this book (REV_DEBT 6,559 to under 9,837 / Broker) is noise. The test data plants nothing in RANR.

Step 12: Read Where it bleeds

Open the workbook again. **Where it bleeds** lists every pocket that loses more than its share, largest first within each measure.

Where it bleeds

Pockets losing more than their share (RANR, earning less), largest first. The flag compares each pocket with the rest of its band. Red: worse. Amber: worse, but could be luck. "Chance it's luck" is the test's p-value.

Measure	Band column	Band	Segment column	Segment	Loans	Rate	Book rate	Excess	Excess is in	Material	Vs rest of book	Chance it's luck	Vs rest of band	Chance it's luck	Flag (vs the rest of its band)	Smallest gap it could show
Outcome, share of loans	FICO	under 654	CHANNEL	Broker	523	24.47%	7.13%	91 loans	yes	yes	4.14x under 0.0001	2.38x under 0.0001	worse	1.47x or more		
Outcome, share of loans	REV_DEBT	17.187 and over	CHANNEL	Broker	507	22.29%	7.13%	77 loans	yes	yes	3.65x under 0.0001	2.01x under 0.0001	worse	1.48x or more		
Outcome, share of loans	FICO	under 654	ASSET_CLASS	4	392	20.41%	7.13%	52 loans	yes	yes	3.17x under 0.0001	1.54x	0.0378 worse	1.55x or more		
Outcome, share of loans	REV_DEBT	17.187 and over	ASSET_CLASS	4	424	18.63%	7.13%	49 loans	yes	yes	2.87x under 0.0001	1.41x	0.0816 worse, but could be luck	1.53x or more		
Outcome, share of loans	REV_DEBT	17.187 and over	ASSET_CLASS	1	411	14.36%	7.13%	30 loans	yes	yes	2.13x under 0.0001	0.98x	0.9513 in line	1.53x or more		
Outcome, share of loans	FICO	under 654	ASSET_CLASS	3	350	15.14%	7.13%	28 loans	yes	yes	2.24x under 0.0001	1.01x	0.9810 in line	1.58x or more		
Outcome, share of loans	FICO	under 654	ASSET_CLASS	1	420	13.57%	7.13%	27 loans	yes	yes	2.01x	0.0002	0.87x	0.6748 in line	1.53x or more	
Outcome, share of loans	REV_DEBT	17.187 and over	ASSET_CLASS	3	380	14.21%	7.13%	27 loans	yes	yes	2.10x	0.0001	0.96x	0.9318 in line	1.56x or more	
Outcome, share of loans	ORIG_BAL	15,449 to under 26,	CHANNEL	Broker	537	11.73%	7.13%	25 loans	yes	yes	1.73x	0.0058	1.58x	0.0557 worse, but could be luck	1.46x or more	
Outcome, share of loans	REV_DEBT	17.187 and over	CHANNEL	Branch	525	11.62%	7.13%	24 loans	yes	yes	1.71x	0.0014	0.72x	0.0631 better, but could be luck	1.47x or more	
Outcome, share of loans	REV_DEBT	17.187 and over	CHANNEL	Online	568	10.56%	7.13%	20 loans	yes	yes	1.54x	0.0075	0.63x	0.0022 better	1.45x or more	
Outcome, share of loans	ORIG_BAL	37,951 to under 49,	ASSET_CLASS	4	414	11.59%	7.13%	18 loans	yes	yes	1.68x	0.0237	1.79x	0.0639 worse, but could be luck	1.53x or more	
Outcome, share of loans	FICO	under 654	CHANNEL	Online	531	10.36%	7.13%	17 loans	yes	yes	1.50x	0.0180	0.59x	0.0004 better	1.47x or more	
Outcome, share of loans	FICO	654 to under 686	ASSET_CLASS	4	410	11.22%	7.13%	17 loans	yes	yes	1.62x	0.0124	1.56x	0.1245 worse, but could be luck	1.54x or more	
Outcome, share of loans	FICO	under 654	ASSET_CLASS	2	393	11.20%	7.13%	16 loans	yes	yes	1.62x	0.0147	0.68x	0.0897 better, but could be luck	1.55x or more	
Outcome, share of loans	REV_DEBT	13.051 to under 17,	ASSET_CLASS	4	411	10.85%	7.13%	16 loans	yes	yes	1.58x	0.0185	1.51x	0.1515 worse, but could be luck	1.53x or more	
Outcome, share of loans	REV_DEBT	13.051 to under 17,	CHANNEL	Broker	553	9.95%	7.13%	16 loans	yes	yes	1.44x	0.0277	1.37x	0.2228 worse, but could be luck	1.46x or more	
Outcome, share of loans	FICO	under 654	CHANNEL	Branch	501	10.18%	7.13%	15 loans	yes	yes	1.47x	0.0295	0.59x	0.0004 better	1.48x or more	
Outcome, share of loans	REV_DEBT	17.187 and over	ASSET_CLASS	2	385	10.91%	7.13%	15 loans	yes	yes	1.57x	0.0225	0.69x	0.0816 better, but could be luck	1.55x or more	
Outcome, share of loans	ORIG_BAL	15,449 to under 26,	ASSET_CLASS	4	413	10.41%	7.13%	14 loans	yes	yes	1.50x	0.0863	1.25x	0.3981 in line	1.53x or more	
Outcome, share of loans	ORIG_BAL	37,951 to under 49,	CHANNEL	Broker	531	9.23%	7.13%	11 loans	yes	yes	1.32x	0.2076	1.30x	0.2552 worse, but could be luck	1.47x or more	
Outcome, share of loans	ORIG_BAL	26,324 to under 37,	ASSET_CLASS	4	412	9.71%	7.13%	11 loans	yes	yes	1.39x	0.1958	1.80x	0.0707 worse, but could be luck	1.53x or more	
Outcome, share of loans	ORIG_BAL	15,449 to under 26,	ASSET_CLASS	3	427	9.37%	7.13%	10 loans	yes	yes	1.34x	0.2429	1.08x	0.9079 in line	1.52x or more	
Outcome, share of loans	FICO	654 to under 686	CHANNEL	Branch	503	8.95%	7.13%	9 loans	yes	yes	1.08x	0.1686	1.13x	0.8208 in line	1.48x or more	
Outcome, share of loans	ORIG_BAL	15,449 to under 26,	ASSET_CLASS	1	404	9.16%	7.13%	8 loans	yes	yes	1.31x	0.2791	1.04x	0.9773 in line	1.54x or more	
Outcome, share of loans	ORIG_BAL	49,152 and over	CHANNEL	Broker	520	8.46%	7.13%	7 loans	yes	yes	1.20x	0.4409	1.90x	0.0517 worse, but could be luck	1.47x or more	
Outcome, share of loans	ORIG_BAL	15,449 to under 49,	ASSET_CLASS	3	382	8.90%	7.13%	7 loans	yes	yes	1.26x	0.3589	1.19x	0.9009 in line	1.56x or more	
Outcome, share of loans	ORIG_BAL	under 15,449	CHANNEL	Broker	514	8.37%	7.13%	6 loans	yes	yes	1.19x	0.4479	1.42x	0.2005 worse, but could be luck	1.47x or more	
Outcome, share of loans	FICO	654 to under 686	CHANNEL	Broker	531	8.10%	7.13%	5 loans	below the line	yes	1.15x	0.4255	0.97x	0.3828 in line	1.47x or more	
Outcome, share of loans	ORIG_BAL	under 15,449	ASSET_CLASS	4	425	8.24%	7.13%	5 loans	below the line	yes	1.17x	0.5160	1.34x	0.3383 worse, but could be luck	1.53x or more	

Measure	Band column	Band	Segment column	Segment	Loans	Rate	Book rate	Excess	Excess is in	Material	Vs rest of book	Chance it's luck	Vs rest of band	Chance it's luck	Flag (vs the rest of its band)	Smallest gap it could show
Outcome, share of loans	FICO	under 654	CHANNEL	Broker	523	24.47%	7.13%	91 loans		yes	4.14x under 0.0001	2.38x under 0.0001	worse			1.47x or more
Outcome, share of loans	REV_DEBT	17.187 and over	CHANNEL	Broker	507	22.29%	7.13%	77 loans		yes	3.65x under 0.0001	2.01x under 0.0001	worse			1.48x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS	4	392	20.41%	7.13%	52 loans		yes	3.17x under 0.0001	1.54x	0.0378 worse			1.55x or more
Outcome, share of loans	REV_DEBT	17.187 and over	ASSET_CLASS	4	424	18.63%	7.13%	49 loans		yes	2.87x under 0.0001	1.41x	0.0816 worse,		but could be luck	1.53x or more
Outcome, share of loans	REV_DEBT	17.187 and over	ASSET_CLASS	1	411	14.36%	7.13%	30 loans		yes	2.13x under 0.0001	0.98x	0.9513 in line			1.53x or more

Right: the first row is FICO under 654 / CHANNEL Broker: 523 loans, 24.47% bad against a book of 7.13%, an excess of 91 loans, 4.14x the rest of the book, 2.38x the rest of its band, flagged **worse**.

How to read a row:

- **Red rows** are worse, and the test says it isn't luck. **Amber** rows are worse but could be luck.
- **Excess is in** gives the unit: loans for the bad-loan share, dollars of a named column otherwise.
- **The flag** uses the comparison you chose in step 9 (here, the rest of its band). The heading says which.
- **Material** says whether the excess clears your line. Below the line rows are still listed.
- **RANR** rows are shortfalls: pockets earning less than their share. A multiple under 1 is the bad direction.

Watch out: long band labels are cut off in the Band column ("15,449 to under 26,"). Widen the column to read them.

Step 13: Losses vs revenue

Losses vs revenue puts every pocket of at least 30 loans in one of four boxes, by whether its GCO rate is above or below the book's and whether its RANR rate is above or below the book's.

Every pocket's GCO rate and RAR rate, each against the book. Right of 1.00x loses more than the book; above 1.00x earns more. The flags say whether each difference is more than luck. Pockets under the minimum loans are left off.

Chart guide lines (for two 1.00s lead)	1.00	2.27
	1.00	3.96
	0.10	1.00
	3.96	1.00

Band column	Band	Segment column	Segment	Loans	GCO vs book	RANR vs book	Which box	GCO excess (\$) RANR vs book (\$) GCO flag	RANR flag	
REV DEBT	17,187 and over	ASSET_CLASS	4	424	2.90x	0.90x	Losing more, earning less	935,522	-4,160 worse, but could be luck	in line
FICO	under 654	ASSET_CLASS	3	350	2.08x	0.85x	Losing more, earning less	490,554	-5,581 in line	worse, but could be luck
REV DEBT	17,187 and over	CHANNEL	Branch	525	1.71x	0.88x	Losing more, earning less	459,356	-6,390 better, but could be luck	in line
ORIG_BAL	37,951 to under 49,15	ASSET_CLASS	4	144	1.64x	0.72x	Losing more, earning less	446,768	-16,240 worse, but could be luck	in line
REV DEBT	17,187 and over	CHANNEL	Online	568	1.61x	1.00x	Losing more, earning less	421,480	-22 better, but could be luck	in line
REV DEBT	17,187 and over	ASSET_CLASS	1	411	1.85x	0.96x	Losing more, earning less	420,823	-1,782 in line	in line
FICO	654 to under 686	ASSET_CLASS	4	410	1.64x	0.89x	Losing more, earning less	336,769	-4,768 worse, but could be luck	in line
ORIG_BAL	49,152 and over	CHANNEL	Broker	520	1.27x	0.48x	Losing more, earning less	301,083	-47,308 worse	worse, but could be luck
ORIG_BAL	37,951 to under 49,15	CHANNEL	Broker	531	1.04x	0.91x	Losing more, earning less	293,859	-13,962 worse, but could be luck	in line
REV DEBT	17,187 and over	ASSET_CLASS	Branch	385	1.57x	0.98x	Losing more, earning less	275,328	-870 better, but could be luck	in line
FICO	under 654	CHANNEL	Branch	502	1.40x	0.99x	Losing more, earning less	249,020	-584 better	in line
FICO	654 to under 686	CHANNEL	Branch	503	1.30x	0.95x	Losing more, earning less	185,594	-2,454 in line	in line
ORIG_BAL	26,324 to under 37,95	ASSET_CLASS	4	412	1.36x	0.78x	Losing more, earning less	182,458	-9,144 worse, but could be luck	worse, but could be luck
ORIG_BAL	37,951 to under 49,15	ASSET_CLASS	3	382	1.27x	0.84x	Losing more, earning less	171,722	-8,405 in line	in line
ORIG_BAL	37,951 to under 49,15	CHANNEL	Branch	525	1.13x	0.74x	Losing more, earning less	112,272	-19,357 in line	in line
ORIG_BAL	49,152 and over	ASSET_CLASS	4	384	1.09x	0.57x	Losing more, earning less	71,343	-29,342 worse, but could be luck	in line
FICO	654 to under 686	CHANNEL	Broker	531	1.10x	0.90x	Losing more, earning less	63,685	-5,466 in line	in line
ORIG_BAL	26,324 to under 37,95	CHANNEL	Broker	553	1.05x	0.99x	Losing more, earning less	35,857	-465 in line	in line
REV DEBT	under 654	CHANNEL	Broker	523	3.05x	1.05x	Losing more, earning more	1,757,039	2,623 worse	in line
REV DEBT	17,187 and over	CHANNEL	Branch	507	3.05x	1.02x	Losing more, earning more	1,712,421	1,172 worse	in line
FICO	under 654	ASSET_CLASS	4	392	3.19x	1.04x	Losing more, earning more	1,039,185	1,449 worse	in line
REV DEBT	17,187 and over	ASSET_CLASS	3	380	2.12x	1.04x	Losing more, earning more	521,584	1,572 in line	in line
FICO	under 654	ASSET_CLASS	1	420	1.91x	1.03x	Losing more, earning more	471,014	1,275 in line	in line
REV DEBT	13,051 to under 17,18	CHANNEL	Broker	553	1.57x	1.03x	Losing more, earning more	396,992	1,462 worse, but could be luck	in line

Right: the count table at the top right reads 18, 24, 38 and 32 pockets. The rows are sorted by box, then by GCO excess.

Read the flags before the box. The box is decided by which side of 1.00x a pocket falls, and nothing else. The planted pocket (ringed) is in **Losing more, earning more** because its RANR is 1.05x the book's, and its RANR flag says in line: the extra revenue is not a real difference. Treat a box as settled only when that side's flag says worse or better (defect 1).

The flags compare with the rest of the band, not the book. So a row can sit in Losing more with a GCO flag of better (FICO under 654 / Branch, for example) (defect 5).

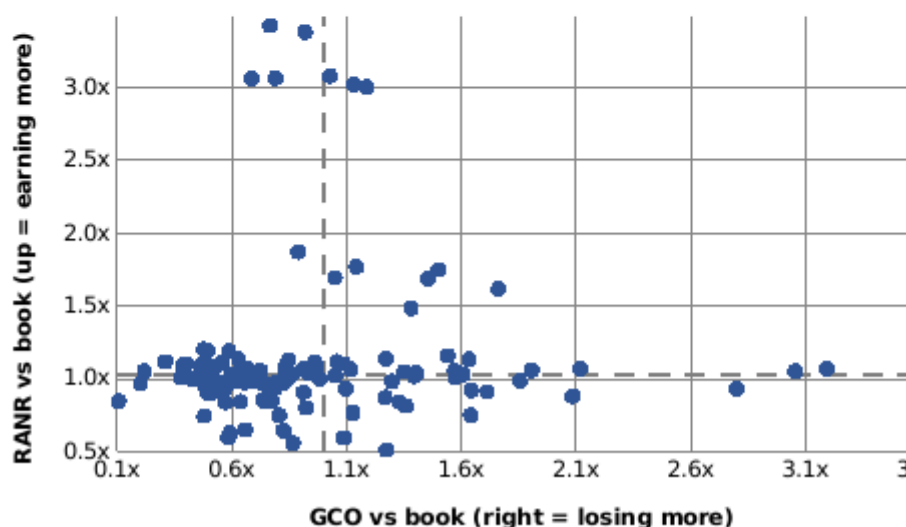
The chart plots the same pockets, GCO across and RANR up, with a dashed line at 1.00x on each:

Box	Pockets
Losing more, earning less	18
Losing more, earning more	24
Losing less, earning less	38
Losing less, earning more	32

Chart guide lines (the two 1.00x lines)

1.00	0.47
1.00	3.46
0.10	1.00
3.86	1.00

Each pocket: GCO against RANR, both vs the book



The dots aren't labelled. Use the table to find a pocket.

Step 14: Grids

Grids shows each band crossed with each segment, one block per measure: the rate, the rate against the book, and the rate against the rest of the same band. Red is worse and green is better.

Grids

Each grid shows ways: the rate, the rate against the book, and the rate against the rest of the same band. Red is worse, green is better. For RANR more is better, so low is red.

FICO x CHANNEL: Outcome, share of loans

Shows where BAD PLAS is 2, and a share of all loans

Rate	Branch	Broker	Online	All
under 654	10.18%	24.47%	10.36%	15.05%
654 to under 686	8.95%	8.10%	7.72%	8.25%
686 to under 712	5.44%	5.19%	4.21%	4.93%
712 to under 746	4.09%	3.14%	3.90%	3.72%
746 and over	4.44%	3.73%	3.68%	3.95%
(blank)		0.00%		0.00%
(marked missing)	1.96%	12.96%	3.64%	6.25%
All	6.47%	9.04%	5.89%	7.13%

FICO x CHANNEL: Outcome, share of booked dollars

Shows where BAD PLAS is 2, and a share of all booked dollars

Rate	Branch	Broker	Online	All
under 654	10.18%	24.47%	10.36%	15.05%
654 to under 686	8.95%	8.10%	7.72%	8.25%
686 to under 712	5.44%	5.19%	4.21%	4.93%
712 to under 746	4.09%	3.14%	3.90%	3.72%
746 and over	4.44%	3.73%	3.68%	3.95%
(blank)		0.00%		0.00%
(marked missing)	1.96%	12.96%	3.64%	6.25%
All	6.47%	9.04%	5.89%	7.13%

FICO x CHANNEL: RANR per booked dollar

Shows where BAD PLAS is 2, and a share of all booked dollars

Rate	Branch	Broker	Online	All
under 654	0.32%	0.34%	0.32%	0.33%
654 to under 686	0.31%	0.29%	0.34%	0.31%
686 to under 712	0.31%	0.31%	0.30%	0.30%
712 to under 746	0.32%	0.29%	0.35%	0.32%
746 and over	0.31%	0.34%	0.32%	0.33%
(blank)		0.20%		0.20%
(marked missing)	0.30%	0.33%	0.23%	0.28%
All	0.32%	0.31%	0.33%	0.32%

FICO x CHANNEL: RANR per booked dollar (more is better)

Shows where BAD PLAS is 2, and a share of all booked dollars

Rate	Branch	Broker	Online	All
under 654	0.32%	0.34%	0.32%	0.33%
654 to under 686	0.31%	0.29%	0.34%	0.31%
686 to under 712	0.31%	0.31%	0.30%	0.30%
712 to under 746	0.32%	0.29%	0.35%	0.32%
746 and over	0.31%	0.34%	0.32%	0.33%
(blank)		0.20%		0.20%
(marked missing)	0.30%	0.33%	0.23%	0.28%
All	0.32%	0.31%	0.33%	0.32%

Vs the book	Branch	Broker	Online	All
under 654	0.91x	1.00x	1.01x	1.00x
654 to under 686	1.26x	1.14x	1.07x	1.16x
686 to under 712	0.76x	0.73x	0.59x	0.69x
712 to under 746	0.57x	0.44x	0.55x	0.52x
746 and over	0.62x	0.52x	0.52x	0.55x
(blank)		0.00x		0.00x
(marked missing)	0.28x	1.82x	0.51x	0.88x
All	0.91x	1.27x	0.83x	1.00x

Vs the book	Branch	Broker	Online	All
under 654	0.91x	1.00x	1.01x	1.00x
654 to under 686	1.26x	1.14x	1.07x	1.16x
686 to under 712	0.76x	0.73x	0.59x	0.69x
712 to under 746	0.57x	0.44x	0.55x	0.52x
746 and over	0.62x	0.52x	0.52x	0.55x
(blank)		0.00x		0.00x
(marked missing)	0.28x	1.82x	0.51x	0.88x
All	0.91x	1.27x	0.83x	1.00x

Vs the book	Branch	Broker	Online	All
under 654	0.91x	1.00x	1.01x	1.00x
654 to under 686	1.26x	1.14x	1.07x	1.16x
686 to under 712	0.76x	0.73x	0.59x	0.69x
712 to under 746	0.57x	0.44x	0.55x	0.52x
746 and over	0.62x	0.52x	0.52x	0.55x
(blank)		0.00x		0.00x
(marked missing)	0.28x	1.82x	0.51x	0.88x
All	0.91x	1.27x	0.83x	1.00x

Vs the book	Branch	Broker	Online	All
under 654	0.91x	1.00x	1.01x	1.00x
654 to under 686	0.95x	0.90x	1.07x	0.97x
686 to under 712	0.95x	0.95x	0.95x	0.95x
712 to under 746	1.01x	0.88x	1.08x	1.00x
746 and over	1.16x	1.07x	1.02x	1.08x
(blank)		0.61x		0.61x
(marked missing)	0.94x	1.02x	0.72x	0.88x
All	1.01x	0.97x	1.02x	1.00x

Vs the rest of its b	Branch	Broker	Online
under 654	0.91x	2.38x	0.91x
654 to under 686	1.13x	0.97x	0.91x
686 to under 712	1.16x	1.08x	0.79x
712 to under 746	1.15x	0.79x	1.07x
746 and over	1.20x	0.92x	0.90x
(blank)		0.00x	
(marked missing)	0.24x	4.58x	0.48x
All			

Vs the rest of its b	Branch	Broker	Online
under 654	0.91x	2.38x	0.91x
654 to under 686	1.13x	0.97x	0.91x
686 to under 712	1.16x	1.08x	0.79x
712 to under 746	1.15x	0.79x	1.07x
746 and over	1.20x	0.92x	0.90x
(blank)		0.00x	
(marked missing)	0.24x	4.58x	0.48x
All			

Vs the rest of its b	Branch	Broker	Online
under 654	0.91x	2.38x	0.91x
654 to under 686	1.13x	0.97x	0.91x
686 to under 712	1.16x	1.08x	0.79x
712 to under 746	1.15x	0.79x	1.07x
746 and over	1.20x	0.92x	0.90x
(blank)		0.00x	
(marked missing)	0.24x	4.58x	0.48x
All			

Vs the rest of its b	Branch	Broker	Online
under 654	0.91x	2.38x	0.91x
654 to under 686	0.97x	0.88x	1.16x
686 to under 712	1.00x	1.00x	1.00x
712 to under 746	1.01x	0.88x	1.14x
746 and over	1.12x	0.99x	0.90x
(blank)		0.00x	
(marked missing)	1.09x	1.24x	0.73x
All			

FICO x CHANNEL: Outcome, share of loans

Shows where BAD PLAS is 2, and a share of all loans

Rate	Branch	Broker	Online	All
under 654	10.18%	24.47%	10.36%	15.05%
654 to under 686	8.95%	8.10%	7.72%	8.25%
686 to under 712	5.44%	5.19%	4.21%	4.93%
712 to under 746	4.09%	3.14%	3.90%	3.72%
746 and over	4.44%	3.73%	3.68%	3.95%
(blank)		0.00%		0.00%
(marked missing)	1.96%	12.96%	3.64%	6.25%
All	6.47%	9.04%	5.89%	7.13%

Vs the book	Branch	Broker	Online	All
under 654	1.43x	3.43x	1.45x	2.11x
654 to under 686	1.26x	1.14x	1.08x	1.16x
686 to under 712	0.76x	0.73x	0.59x	0.69x
712 to under 746	0.57x	0.44x	0.55x	0.52x
746 and over	0.62x	0.52x	0.52x	0.55x
(blank)		0.00x		0.00x
(marked missing)	0.28x	1.82x	0.51x	0.88x
All	0.91x	1.27x	0.83x	1.00x

Vs the rest of its b	Branch	Broker	Online
under 654	0.59x	2.38x	0.59x
654 to under 686	1.13x	0.97x	0.91x
686 to under 712	1.16x	1.08x	0.79x
712 to under 746	1.15x	0.79x	1.07x
746 and over	1.20x	0.92x	0.90x
(blank)		0.00x	
(marked missing)	0.24x	4.58x	0.48x
All			

Right: in FICO x CHANNEL: Outcome, share of loans, under 654 / Broker is 24.47%, 3.43x the book and 2.38x the rest of its band.

For RANR, **more is better**, and the heading says so. Low is red:

All	3.40%	5.02%	3.16%	3.86%
All	0.88x	1.30x	0.82x	1.00x
All				

FICO x CHANNEL: RANR per booked dollar (more is better)	Branch	Broker	Online	All
under 654	0.32%	0.34%	0.32%	0.33%
654 to under 686	0.31%	0.29%	0.34%	0.31%
686 to under 712	0.31%	0.31%	0.30%	0.30%
712 to under 746	0.32%	0.29%	0.35%	0.32%
746 and over	0.31%	0.34%	0.32%	0.33%
(blank)		0.20%		0.20%
(marked missing)	0.30%	0.33%	0.23%	0.28%
All	0.32%	0.31%	0.33%	0.32%

Vs the book	Branch	Broker	Online	All
under 654	0.99x	1.05x	1.01x	1.02x
654 to under 686	0.95x	0.90x	1.07x	0.97x
686 to under 712	0.95x	0.95x	0.95x	0.95x
712 to under 746	1.01x	0.89x	1.08x	1.00x
746 and over	1.16x	1.07x	1.02x	1.08x
(blank)		0.61x		0.61x
(marked missing)	0.94x	1.02x	0.72x	0.88x
All	1.01x	0.97x	1.02x	1.00x

Vs the rest of its b	Branch	Broker	Online
under 654	0.96x	1.05x	0.99x
654 to under 686	0.97x	0.89x	1.16x
686 to under 712	1.00x	1.00x	1.00x
712 to under 746	1.01x	0.88x	1.14x
746 and over	1.12x	0.99x	0.90x
(blank)		0.00x	
(marked missing)	1.09x	1.24x	0.73x
All			

Watch out: the grids colour every cell, tested or not. The (blank) row holds one loan, and (marked missing) Broker holds 54 loans with too few losses to test. Both show some of the strongest colours on the page (defect 4).

Step 15: Materiality

Materiality shows, for every grid and measure, what each level would keep: how many pockets and how much of the grid's excess.

Materiality

What each materiality level would keep: how many pockets, and how much of the grid's excess they hold. The level itself is set on Control.

FICO x CHANNEL: Outcome, share of loans

In use: 5.7 loans

Share of the book's total	Excess at or over (loans)	Pockets kept	Share of the grid's excess
0.5%	2.9	7	100%
1.0%	5.7	4	92%
2.0%	11.4	3	86%
5.0%	28.5	1	63%
10.0%	57.0	1	63%

FICO x CHANNEL: Outcome, share of booked dollars

In use: 178,222 ORIG_BAL dollars

Share of the book's total	Excess at or over (ORIG_BAL dollars)	Pockets kept	Share of the grid's excess
0.5%	89,111	5	97%
1.0%	178,222	4	94%
2.0%	356,445	3	88%
5.0%	891,112	1	65%
10.0%	1,782,225	1	65%

FICO x CHANNEL: GCO per booked dollar

In use: 99,554 GCO_AMT dollars

Share of the book's total	Excess at or over (GCO_AMT dollars)	Pockets kept	Share of the grid's excess
0.5%	49,777	6	99%
1.0%	99,554	4	95%
2.0%	199,107	3	87%

Right: for FICO x CHANNEL: Outcome, share of loans, the level in use is 5.7 loans (1% of the book's bad loans), which keeps 4 pockets and 92% of the excess. Going to 2% keeps 3 pockets and 86%.

Only 1% and 5% can be picked on Control. For another level, work out its dollars from this tab and type them in Or enter your own.

Step 16: Check

Check records what the run did: the extract and loan count, the record file, the tie-outs, what was left out and why, the band edges, the loans needed to show a 1.25x gap, each materiality line with its unit, and every setting in words.

Check

Settings used, tie-outs, and what was left out.

Extract	Consumer book Q3.csv
Loans run	8,000
Record of this run	Consumer book Q3 - Origination Cube - what ran.yaml, beside this workbook: every setting the run used, kept for the file. It is replaced by the next run.
Tie-out checks	132 of 132 agree: every grid adds up to the book
Left out of Outcome, share of loans	1 x BAD_FLAG flag not 0 or 1
Left out of Outcome, share of booked dollars	1 x ORIG_BAL blank; 1 x BAD_FLAG flag not 0 or 1
Left out of GCO per booked dollar	1 x GCO_AMT not a number; 1 x ORIG_BAL blank
Left out of RANR per booked dollar	1 x ORIG_BAL blank
Band edges used: FICO	654; 686; 712; 746 (5 bands)
Band edges used: ORIG_BAL	15,449; 26,324; 37,951; 49,152 (5 bands)
Band edges used: REV_DEBT	6,559; 9,837; 13,051; 17,187 (5 bands)
Loans needed for a 1.25x gap: Outcome, share of loans	about 1,756
Loans needed for a 1.25x gap: Outcome, share of booked dollars	about 2,203
Loans needed for a 1.25x gap: GCO per booked dollar	about 2,359
Loans needed for a 1.25x gap: RANR per booked dollar	about 411
Materiality line: Outcome, share of loans	5.7 loans
Materiality line: Outcome, share of booked dollars	178,222 ORIG_BAL dollars
Materiality line: GCO per booked dollar	99,554 GCO_AMT dollars
Materiality line: RANR per booked dollar	8,288 RANR_AMT dollars
How old a loan must be to count	Every loan
Fewest loans in a pocket before it is tested	30 loans
Fewest loans with a loss before a loss rate is tested	10 losses
Smallest excess loss worth reporting	1% of the book's total losses
What a pocket is judged against	The rest of its band
How much worse than its comparison a pocket must be	1.25 times
How much better than its comparison a pocket must be	0.8 times
How sure a difference must be before it counts	95% sure
How often a real gap should be caught	80% of the time
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg)

Right: "132 of 132 agree", five bands for each number column, and the settings match step 9.

Step 17: Log, and Start here

Log keeps every run, newest first, including refused ones.

Log

Every Run and every refusal, newest first.

2026-09-25 16:46

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

2026-09-25 16:45

Couldn't run. Fix these, save, close, and press Run again:
Control!C6: "How old a loan must be to count" needs an answer. Pick one, or enter your own in column D.
Control!C13: "Fewest loans in a pocket before it is tested" needs an answer. Pick one, or enter your own in column D.
Control!C14: "Fewest loans with a loss before a loss rate is tested" needs an answer. Pick one, or enter your own in column D.

Control!C16: "Smallest excess loss worth reporting" needs an answer. Pick one, or enter your own in column D.
Control!C18: "What a pocket is judged against" needs an answer. Pick one from the list.
Control!C19: "How much worse than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
Control!C20: "How much better than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
Control!C22: "How sure a difference must be before it counts" needs an answer. Pick one, or enter your own in column D.

Columns!C3: set Checked every column to Yes once you've checked each column's meaning.

Start here now shows 0 calls left and the time of the last run:

Origination Cube

Set up from Consumer book Q3.csv: 8,000 loans, 9 columns.

Step	Where	What to do
1	Control	Fill in the shaded cells: materiality, minimum loans, how much worse counts.
2	Columns	Check each column's meaning and whether to cut by it. Set C3 to Yes when done.
3	Odd values	Answer real or missing. Unanswered ones are used as is.
4	Launcher	Save and close this workbook, then press Run the cube.

Where things stand

Calls still to make on Control

0

Things to look at first on Columns

7

Odd values found

2

Last run

2026-09-25 16:46: Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

Step 18: Split every pocket by a number (revolving debt)

On **Columns**, set **Split pockets by it?** to **Yes** on one number column. The walk used **REV_DEBT**. Save, close, and Run.

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done, confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (use ; between)	Show per pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)								
FICO	FICO score	Yes				160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.	name contains 'loan'; a different value on every row	0%	L0000000, L0000001, L0000002
CHANNEL	Category	Yes				suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.	name contains 'fico'; 98% of values between 300 and 850	0%	-9999, 713, 716
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes				1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	name contains 'bal'; positive on every loan	0%	7655.75, 27999.47, 39684.43
GCO_AMT	GCO dollars					1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	name contains 'bad'; 0 and 1 (7.1% are 1)	0%	0, 2, 1
RANR_AMT	RANR dollars					2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.	name contains 'gco'; zero on 93% of loans, never negative	0%	0.0, #N/A, 27620.62
ASSET_CLASS	Category	Yes				suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.	name contains 'ranr'; numbers	0%	-165.2, 54.71, 38.01
REV_DEBT	Amount or number	Yes			Yes	suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.	4 different values	0%	1, 2, 4
							numbers with 7,889 values	0%	15868.86, 12884.7, 8920.97

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (use ; between)	Show per pocket	Split pockets by it?	Look first
LOAN_NBR	Loan number (key)					
FICO	FICO score	Yes				160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.
CHANNEL	Category	Yes				suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes				1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.
GCO_AMT	GCO dollars					1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.
RANR_AMT	RANR dollars					2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.
ASSET_CLASS	Category	Yes				suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.
REV_DEBT	Amount or number	Yes			Yes	suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.

Origination Cube

Extract: /tmp/claude-0/-home-user-SATC/b4305bfc-8769-5da1-bd67-

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 493 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Right: "493 tie-out checks agree". The window doesn't mention the split. Open the new Split tab.

Split

Every pocket split in two at its own median REV_DEBT: the high half against the low half, holding the band and segment fixed. Above 1.05x means the high half does worse (for RANR, earns more). REV_DEBT isn't cut into bands of its own while it splits.

FICO x CHANNEL: Outcome, share of loans

The high-REV_DEBT half was worse in 16 of 18 pockets. Pooled across pockets, its odds of the outcome are 2.13x the low half's (95% range 1.77 to 2.55); chance it's luck under 0.0001. The effect is about the same in every pocket (p 0.3086).

High half vs low	Branch	Broker	Online
under 654	2.39x	2.67x	1.63x
654 to under 686	1.37x	2.92x	2.26x
686 to under 712	2.12x	1.90x	1.56x
712 to under 746	2.14x	0.78x	1.44x
746 and over	1.30x	2.33x	1.12x
(blank)			
(marked missing)		6.00x	1.04x

Chance it's luck	Branch	Broker	Online
under 654	0.0018	under 0.0001	0.0618
654 to under 686	0.2686	0.0007	0.0117
686 to under 712	0.0504	0.1048	0.2871
712 to under 746	0.0813	0.6176	0.3849
746 and over	0.5229	0.0679	0.8089
(blank)			
(marked missing)	0.3173	0.0386	0.9795

FICO x CHANNEL: Outcome, share of booked dollars

The high-REV_DEBT half was worse in 16 of 18 pockets. Holding the pocket fixed, the high half loses 2.08x what it would at the low half's rate; chance it's luck under 0.0001.

High half vs low	Branch	Broker	Online
under 654	2.84x	2.11x	2.05x
654 to under 686	1.58x	3.82x	2.83x
686 to under 712	2.39x	2.69x	1.30x
712 to under 746	2.68x	1.12x	1.63x
746 and over	1.17x	2.82x	1.01x
(blank)			
(marked missing)		4.88x	0.19x

Chance it's luck	Branch	Broker	Online
under 654	0.0009	under 0.0001	0.0169
654 to under 686	0.1591	0.0004	0.0044
686 to under 712	0.0512	0.0317	0.5645
712 to under 746	0.0458	0.8406	0.3017
746 and over	0.7324	0.0523	0.9881
(blank)			
(marked missing)	0.3221	0.1058	0.4221

FICO x CHANNEL: GCO per booked dollar

The high-REV_DEBT half was worse in 15 of 18 pockets. Holding the pocket fixed, the high half loses 2.06x what it would at the low half's rate; chance it's luck under 0.0001.

FICO x CHANNEL: Outcome, share of loans

The high-REV_DEBT half was worse in 16 of 18 pockets. Pooled across pockets, its odds of the outcome are 2.13x the low half's (95% range 1.77 to 2.55); chance it's luck under 0.0001. The effect is about the same in every pocket (p 0.3086).

High half vs low	Branch	Broker	Online
under 654	2.39x	2.67x	1.63x
654 to under 686	1.37x	2.92x	2.26x
686 to under 712	2.12x	1.90x	1.56x
712 to under 746	2.14x	0.78x	1.44x
746 and over	1.30x	2.33x	1.12x
(blank)			
(marked missing)		6.00x	1.04x

Chance it's luck	Branch	Broker	Online
under 654	0.0018	under 0.0001	0.0618
654 to under 686	0.2686	0.0007	0.0117
686 to under 712	0.0504	0.1048	0.2871
712 to under 746	0.0813	0.6176	0.3849
746 and over	0.5229	0.0679	0.8089
(blank)			
(marked missing)	0.3173	0.0386	0.9795

How to read it: every pocket is cut in two at its own median revolving debt. High half vs low compares the two halves inside each pocket, and the sentence above each block pools the pockets. For FICO x CHANNEL the high half was worse in 16 of 18 pockets, with odds 2.13x the low half's (95% range 1.77 to 2.55).

Read the FICO grids, not the others. A split only holds fixed what the grid holds fixed. In the ORIG_BAL grids nothing holds the score, so high revolving debt also means lower scores, and the answer comes out at 2.73x and 2.76x:

The high-REV_DEBT half was worse in 20 of 24 pockets. Pooled across pockets, its odds of the outcome are 2.06x the low half's (95% range 1.72 to 2.47); chance it's luck under 0.0001. The effect is about the same in every pocket (p 0.7059).

High half vs low	1	2	3	4
under 654	3.07x	1.92x	1.94x	2.08x
654 to under 686	2.64x	3.22x	1.59x	1.71x
686 to under 712	3.25x	0.72x	2.50x	1.71x
712 to under 746	1.76x	1.15x	0.67x	1.67x
746 and over	0.88x	3.02x	1.68x	1.51x
(blank)				
(marked missing)	4.00x	1.05x	1.05x	

FICO x ASSET_CLASS: Outcome, share of booked dollars

The high-REV_DEBT half was worse in 20 of 24 pockets. Holding the pocket fixed, the high half loses 1.99x what it would at the low half's rate; chance it's luck under 0.0001.

High half vs low	1	2	3	4
under 654	2.17x	1.80x	2.07x	2.11x
654 to under 686	3.35x	5.38x	1.54x	2.36x
686 to under 712	2.92x	0.84x	2.30x	2.29x
712 to under 746	1.81x	1.07x	1.05x	2.35x
746 and over	0.70x	3.73x	1.11x	1.56x
(blank)				
(marked missing)	1.91x	0.20x	1.11x	1.11x

FICO x ASSET_CLASS: GCO per booked dollar

The high-REV_DEBT half was worse in 18 of 24 pockets. Holding the pocket fixed, the high half loses 1.97x what it would at the low half's rate; chance it's luck under 0.0001.

High half vs low	1	2	3	4
under 654	2.07x	1.98x	1.96x	2.05x
654 to under 686	2.89x	5.33x	1.67x	1.97x
686 to under 712	3.89x	0.72x	2.95x	2.31x
712 to under 746	2.28x	0.77x	0.77x	2.27x
746 and over	0.79x	3.08x	1.21x	1.94x
(blank)				
(marked missing)	2.64x	0.20x	1.54x	

FICO x ASSET_CLASS: RANR per booked dollar

The high-REV_DEBT half was worse in 8 of 24 pockets. Holding the pocket fixed, the high half earns 1.06x what it would at the low half's rate; chance it's luck 0.1638.

High half vs low	1	2	3	4
under 654	1.09x	1.01x	0.95x	1.01x
654 to under 686	0.87x	0.88x	1.43x	1.03x
686 to under 712	1.02x	0.98x	0.90x	0.99x
712 to under 746	1.53x	1.12x	1.07x	1.04x
746 and over	1.10x	0.87x	1.07x	1.21x
(blank)				
(marked missing)	2.08x	2.39x	0.75x	2.77x

Chance it's luck	1	2	3	4
under 654	under 0.0001	0.0256	0.0107	0.0004
654 to under 686	0.0113	0.0124	0.1846	0.0599
686 to under 712	0.0249	0.5643	0.0429	0.1596
712 to under 746	0.3545	0.7862	0.5226	0.2037
746 and over	0.7930	0.0399	0.3029	0.4241
(blank)				
(marked missing)	0.3173	0.1477	0.9738	0.9711

Chance it's luck	1	2	3	4
under 654	0.0083	0.0746	0.0144	0.0007
654 to under 686	0.0091	0.0032	0.2547	0.0085
686 to under 712	0.0511	0.7963	0.1006	0.0755
712 to under 746	0.3561	0.9070	0.9412	0.0605
746 and over	0.5329	0.0285	0.8464	0.4444
(blank)				
(marked missing)	0.3342	0.5202	0.4273	0.9407

Chance it's luck	1	2	3	4
under 654	0.0150	0.0487	0.0254	0.0019
654 to under 686	0.0248	0.0050	0.1927	0.0427
686 to under 712	0.0245	0.6200	0.0475	0.0713
712 to under 746	0.2331	0.6712	0.6840	0.0836
746 and over	0.6708	0.0564	0.7355	0.2703
(blank)				
(marked missing)	0.3342	0.3400	0.4271	0.7709

Chance it's luck	1	2	3	4
under 654	0.5966	0.9622	0.7821	0.9737
654 to under 686	0.4910	0.5119	0.0281	0.8761
686 to under 712	0.9100	0.9162	0.5366	0.9705
712 to under 746	0.0251	0.5574	0.6740	0.8129
746 and over	0.5558	0.3808	0.7189	0.2778
(blank)				
(marked missing)	0.1867	0.2215	0.6582	0.0744

ORIG_BAL x CHANNEL: Outcome, share of loans

The high-REV_DEBT half was worse in 15 of 15 pockets. Pooled across pockets, its odds of the outcome are 2.73x the low half's (95% range 2.26 to 3.29); chance it's luck under 0.0001. The effect is about the same in every pocket (p 0.1117).

High half vs low	Booked	Booked	Online
under 15,449	1.92x	4.38x	1.90x
15,449 to under 26.3	1.47x	2.72x	1.63x
26,324 to under 37.9	1.76x	5.85x	5.00x
37,951 to under 49.1	3.67x	2.51x	2.41x

Chance it's luck	Booked	Booked	Online
under 15,449	0.0542	under 0.0001	0.0856
15,449 to under 26.3	0.2284	under 0.0001	0.1079
26,324 to under 37.9	0.1029	under 0.0001	0.0001
37,951 to under 49.1	under 0.0001	0.0015	0.0125

On a copy of this book with **no** revolving-debt effect at all, the ORIG_BAL grids still said 1.66x and 1.70x, "chance it's luck under 0.0001" (defect 2). The planted effect is 1.8x the bad rate. The FICO grids read 1.97x and 1.93x as a bad-rate ratio (2.13x and 2.06x as odds).

REV_DEBT isn't cut into bands of its own while it splits, and the tab's subtitle says so.

Step 19: Split every pocket by a category (asset class)

Clear REV_DEBT's split and set **ASSET_CLASS** instead. Save, close, and Run.

Origination Cube

Extract: /tmp/claude-0/-home-user-SATC/b4305bfc-8769-5da1-bd67-

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 316 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx; start with Where it bleeds.

Right: "316 tie-out checks agree". On **Grids**, every block is followed by one small grid per asset class, each showing the rate against the book:

Grids

Each grid three ways: the rate, the rate against the book, and the rate against the rest of the same book. Red is worse, green is better. For ASSET more is better, so low is red.

FICO x CHANNEL: Outcome, share of loans

Rate	Branch	Broker	Online	All
under 654	10.31%	24.47%	10.30%	15.25%
654 to under 686	8.93%	8.20%	7.72%	8.25%
686 to under 712	5.44%	5.19%	4.21%	4.93%
712 to under 746	4.09%	3.24%	3.40%	3.25%
746 and over	4.44%	3.73%	3.69%	3.95%
(blank)				0.00%
(marked missing)	1.96%	12.86%	3.64%	0.00%
All	6.47%	9.04%	5.09%	7.17%

Vs the book	Branch	Broker	Online	All
under 654	1.41x	0.68x	1.45x	2.14x
654 to under 686	1.26x	1.14x	1.08x	1.16x
686 to under 712	0.76x	0.73x	0.59x	0.69x
712 to under 746	0.85x	0.84x	0.55x	0.58x
746 and over	0.62x	0.52x	0.52x	0.53x
(blank)		1.06x	0.96x	0.96x
(marked missing)	0.28x	1.42x	0.55x	0.88x
All	0.81x	1.27x	0.83x	1.00x

Vs the rest of its b	Branch	Broker	Online	All
under 654	0.57x	2.41x	0.57x	0.56x
654 to under 686	1.13x	0.97x	0.91x	0.91x
686 to under 712	1.36x	1.08x	0.79x	0.79x
712 to under 746	1.15x	0.78x	1.07x	1.07x
746 and over	1.20x	0.92x	0.90x	0.90x
(blank)				
(marked missing)	0.74x	4.54x	0.74x	0.74x
All				

...split by ASSET CLASS: the rate against the book, one grid per value

ASSET CLASS = 1	Branch	Broker	Online	All
under 654	1.07x	2.51x	1.07x	1.07x
654 to under 686	1.18x	1.14x	0.76x	0.76x
686 to under 712	0.59x	0.90x	0.48x	0.48x
712 to under 746	0.44x	0.52x	0.33x	0.33x
746 and over	0.70x	0.31x	0.51x	0.51x
(blank)				
(marked missing)	0.00x	1.08x	0.00x	0.00x

ASSET CLASS = 2	Branch	Broker	Online	All
under 654	1.07x	2.51x	1.07x	1.07x
654 to under 686	1.14x	0.34x	1.04x	1.04x
686 to under 712	0.52x	0.37x	0.38x	0.38x
712 to under 746	0.44x	0.52x	0.49x	0.49x
746 and over	0.33x	0.80x	0.54x	0.54x
(blank)		0.00x		
(marked missing)	1.28x	3.24x	1.17x	1.17x

ASSET CLASS = 3	Branch	Broker	Online	All
under 654	0.91x	4.12x	0.91x	1.45x
654 to under 686	0.94x	1.10x	1.41x	1.41x
686 to under 712	0.77x	0.59x	0.96x	0.96x
712 to under 746	0.51x	0.25x	0.29x	0.29x
746 and over	0.58x	0.74x	0.43x	0.43x
(blank)				
(marked missing)	0.00x	0.94x	1.00x	1.00x

FICO x CHANNEL: Outcome, share of booked dollars

Rate	Branch	Broker	Online	All
under 654	2.27x	4.07x	2.25x	2.25x
654 to under 686	3.49x	1.88x	1.12x	1.12x
686 to under 712	1.15x	1.04x	0.61x	0.61x
712 to under 746	0.87x	0.44x	1.02x	1.02x
746 and over	0.88x	0.22x	0.60x	0.60x
(blank)				
(marked missing)	0.00x	2.16x	0.00x	0.00x

Vs the book	Branch	Broker	Online	All
under 654	1.07x	2.58x	1.01x	1.01x
654 to under 686	1.14x	0.34x	1.04x	1.04x
686 to under 712	0.52x	0.37x	0.38x	0.38x
712 to under 746	0.44x	0.52x	0.49x	0.49x
746 and over	0.33x	0.80x	0.54x	0.54x
(blank)		0.00x		
(marked missing)	1.28x	3.24x	1.17x	1.17x

Vs the rest of its b	Branch	Broker	Online	All
under 654	0.91x	4.12x	0.91x	1.45x
654 to under 686	0.94x	1.10x	1.41x	1.41x
686 to under 712	0.77x	0.59x	0.96x	0.96x
712 to under 746	0.51x	0.25x	0.29x	0.29x
746 and over	0.58x	0.74x	0.43x	0.43x
(blank)				
(marked missing)	0.00x	0.94x	1.00x	1.00x

...split by ASSET CLASS: the rate against the book, one grid per value

ASSET CLASS = 1	Branch	Broker	Online	All
under 654	1.07x	2.58x	1.01x	1.01x
654 to under 686	1.18x	1.14x	0.76x	0.76x
686 to under 712	0.59x	0.90x	0.48x	0.48x
712 to under 746	0.44x	0.52x	0.33x	0.33x
746 and over	0.70x	0.31x	0.51x	0.51x
(blank)				
(marked missing)	0.00x	1.08x	0.00x	0.00x

ASSET CLASS = 2	Branch	Broker	Online	All
under 654	1.07x	2.58x	1.01x	1.01x
654 to under 686	1.14x	0.34x	1.04x	1.04x
686 to under 712	0.52x	0.37x	0.38x	0.38x
712 to under 746	0.44x	0.52x	0.49x	0.49x
746 and over	0.33x	0.80x	0.54x	0.54x
(blank)		0.00x		
(marked missing)	1.28x	3.24x	1.17x	1.17x

ASSET CLASS = 3	Branch	Broker	Online	All
under 654	0.91x	4.12x	0.91x	1.45x
654 to under 686	0.94x	1.10x	1.41x	1.41x
686 to under 712	0.77x	0.59x	0.96x	0.96x
712 to under 746	0.51x	0.25x	0.29x	0.29x
746 and over	0.58x	0.74x	0.43x	0.43x
(blank)				
(marked missing)	0.00x	0.94x	1.00x	1.00x

ASSET CLASS = 4

Branch	Broker	Online	
under 654	2.27x	4.07x	2.25x
654 to under 686	1.69x	1.93x	1.12x
686 to under 712	1.15x	1.04x	0.61x
712 to under 746	0.87x	0.44x	1.02x
746 and over	0.88x	0.22x	0.60x
(blank)			
(marked missing)	0.00x	2.16x	0.00x

Asset class 4 is the reddest, as planted (1.4x the others).

Split

Every pocket split by each value of ASSET_CLASS. The grids are on the Grids tab, one per value.

Watch out:

- **ASSET_CLASS stops being a segment.** Where it bleeds, Losses vs revenue and Materiality lose their ASSET_CLASS grids. Nothing says so (defect 3).
- **The split pockets aren't tested or ranked.** They are only in these small grids, with no loan counts. The (marked missing) pockets hold 11 to 16 loans each.

Step 20: Show a median or average per pocket

On **Columns**, set **Show per pocket** to median or average on any number column. The walk put average on ORIG_BAL and median on REV_DEBT. Save, close, and Run.

FICO x CHANNEL: average ORIG_BAL per pocket

Average	Branch	Broker	Online	All
under 654	32,549.22	32,583.3	32,706.55	32,614.37
654 to under 686	32,159.23	32,621.29	31,389.31	32,066.01
686 to under 712	32,698.63	32,967	32,344.62	32,660.97
712 to under 746	30,750.58	31,887.05	32,568.99	31,746.88
746 and over	31,757.39	31,969.53	32,765.47	32,161.52
(blank)		27,999.47		27,999.47
(marked missing)	34,725.56	30,898.33	33,613.26	33,051.52
All	32,022.08	32,369.91	32,390.12	32,262.59

FICO x CHANNEL: median REV_DEBT per pocket

Median	Branch	Broker	Online	All
under 654	17,209.02	16,276.56	17,082.69	16,904.13
654 to under 686	13,555.63	13,291.2	13,668.67	13,535.03
686 to under 712	12,292.05	12,169.43	12,232.89	12,219.33
712 to under 746	9,952.14	10,430.9	10,658.88	10,271.78
746 and over	7,429.6	6,911.21	7,233.76	7,205.29
(blank)		12,884.7		12,884.7
(marked missing)	9,964.06	10,431.42	10,267.36	10,325.93
All	11,443.27	11,393.55	11,328.21	11,392.84

Right: each grid on **Grids** gets a block per column, average ORIG_BAL per pocket and median REV_DEBT per pocket. These are for reading beside the rates. They aren't tested and don't add up across pockets. The tab doesn't say that.

Step 21: Use the LOB's own cut points

If the LOB cuts scores at 620, 680 and 740, type them in that column's **Band edges** cell with **semicolons**: 620; 680; 740. Save, close, and Run.

Origination Cube

Extract:

Ran on 8,000 loans from Consumer book Q3.csv: 132 tie-out checks agree
Worst for Outcome, share of loans: FICO under 620 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 620 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 620 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Right: the worst pocket is now **FICO under 620 / CHANNEL Broker:** 176 loans, 53.4% bad, 8.78x the rest of the book. Check says "620; 680; 740 (4 bands)".

Commas also work if the cell is left as text. If Excel turns the list into one number, Run refuses and names the cell (wrong turn D).

Step 22: The next extract: press Set up again

Press **1. Set up from this extract** again, on the same extract or a refreshed one with the same name.

Origination Cube

Extract: /tmp/claude-0/-home-user-SATC/b4305bfc-8769-5da1-bd67-

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv:
8,000 loans, 9 columns.
2 things to look at first, on the Columns tab.
Everything is answered. Press Run the cube.

Right: "Everything is answered. Press Run the cube." Your Control, Columns and Odd values answers are kept. So are the last run's results and the Log:

Origination Cube

Set up from Consumer book Q3.csv: 8,000 loans, 9 columns.

Step	Where	What to do
1	Control	Fill in the shaded cells: materiality, minimum loans, how much worse counts.
2	Columns	Check each column's meaning and whether to cut by it. Set C3 to Yes when done.
3	Odd values	Answer real or missing. Unanswered ones are used as is.
4	Launcher	Save and close this workbook, then press Run the cube.

Where things stand

Calls still to make on Control	0
Things to look at first on Columns	2
Odd values found	2
Last run	2026-09-25 16:50: Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

If the new extract has columns the old one didn't, C3 is cleared and the new columns are shaded "New since the last check" (wrong turn N).

Step 23: Prune something learned

On **Learned**, set a row's **Keep?** to **Forget**, then save, close and Run.

Learned						
Meanings and answers carried over from past runs. Set a row to Forget to drop it on the next Run. To stop it being learned again, also change it on Columns.						
Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
Keep	column	ASSET_CLASS	Category	2026-09-25	2026-09-25	5
Keep	column	BAD_FLAG	Outcome (yes/no)	2026-09-25	2026-09-25	5
Forget	column	CHANNEL	Category	2026-09-25	2026-09-25	5
Keep	column	FICO	FICO score	2026-09-25	2026-09-25	5
Keep	column	GCO_AMT	GCO dollars	2026-09-25	2026-09-25	5
Keep	column	LOAN_NBR	Loan number (key)	2026-09-25	2026-09-25	5
Keep	column	ORIG_BAL	Booked amount	2026-09-25	2026-09-25	5
Keep	column	RANR_AMT	RANR dollars	2026-09-25	2026-09-25	5
Keep	column	REV_DEBT	Amount or number	2026-09-25	2026-09-25	5
Keep	column	FICO	Amount or number	2026-09-25	2026-09-25	5

Right: the window says "Forgot CHANNEL, as marked on Learned."

Origination Cube

Extract:

1. Set up from this extract 2. Run the cube Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 620 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 620 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 620 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Forgot CHANNEL, as marked on Learned.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Watch out: the next Run learns it again, silently, because Columns still says CHANNEL is a category and C3 still says Yes. After a second run, CHANNEL is back with Times 1 (defect 9):

Learned						
Meanings and answers carried over from past runs. Set a row to Forget to drop it on the next Run. To stop it being learned again, also change it on Columns.						
Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
Keep	column	ASSET_CLASS	Category	2026-09-25	2026-09-25	7
Keep	column	BAD_FLAG	Outcome (yes/no)	2026-09-25	2026-09-25	7
Keep	column	CHANNEL	Category	2026-09-25	2026-09-25	1
Keep	column	FICO	FICO score	2026-09-25	2026-09-25	7
Keep	column	GCO_AMT	GCO dollars	2026-09-25	2026-09-25	7
Keep	column	LOAN_NBR	Loan number (key)	2026-09-25	2026-09-25	7
Keep	column	ORIG_BAL	Booked amount	2026-09-25	2026-09-25	7
Keep	column	RANR_AMT	RANR dollars	2026-09-25	2026-09-25	7
Keep	column	REV_DEBT	Amount or number	2026-09-25	2026-09-25	7
Keep	column	FICO	Amount or number	2026-09-25	2026-09-25	7

To stop it being learned, change its meaning on Columns as well, as the Learned tab's subtitle says.

When something goes wrong

What you did	What the window says	What to do
A. Split set on two columns	"Columns!H: only one column can split the pockets; ASSET_CLASS, REV_DEBT are all set to Yes."	Clear one.
B. Split the only category that's cut, with the other set to Cut by it? No	"Nothing is left to cut across ... Changed from what was suggested: ASSET_CLASS (Columns! D13, Cut by it? No)." It doesn't say the split took CHANNEL out.	Clear the split, or set ASSET_CLASS back to Yes.
C. average on a category	"Columns!G8: "CHANNEL" isn't a number column, so it has no average to show."	Clear the cell.
D. Edges typed as 620,680,740 and turned into one number by Excel	"Columns!F7: the band edges for "FICO" read as the single number 620,680,740. Excel dropped the commas. Type them with semicolons: 620; 680; 740."	Retype with semicolons.
D2. Edges outside the column's range (620; 680; 950)	"Columns!F7: FICO runs from 496 to 921, so an edge at 950 would leave a band empty."	Type edges inside that range.
E. Workbook still open in Excel	"... is open in Excel. Close it, then press Run again." Set up says the same. Nothing else changes.	Close Excel and press again.
F. A column renamed in the extract after Set up	"Couldn't run: the extract has no column "CHANNEL" (used by dimension channel). Its columns are: ..."	Press Set up, then Run.
G. Rows changed in the extract after Set up	"Consumer book Q3.csv has changed since Set up...", then it runs on the new rows.	Check the loan count.
H. Workbook copied into a new folder with a new extract of the same name	The same "has changed since Set up" line; it runs the new folder's extract (3,000 loans).	Press Set up in the new folder first.
I. Split set on GCO_AMT, or on the loan number	Runs. The Split tab says "worse in 1 of 14 pockets ... loses 93.83x" (GCO), or "Not enough loans in both halves" (loan number).	Split only by something known at booking.
J. A dollar amount typed as your own materiality	Runs. Check carries a warning with the machine name "outcome_loans", and the same dollars become the RANR line.	Prefer the percentage options.
K. The workbook picked as the extract	"Set up Consumer book Q3 - Origination Cube - Origination Cube.xlsx from Consumer book Q3 - Origination Cube.xlsx: 11 loans, 4 columns."	Delete the new file and pick the .csv.
L. 95 typed for "how sure"	"... needs a share, such as 0.95 for 95%, from 0.5 to 0.999; got 95. For 95%, type 0.95."	Type 0.95, or pick "95% sure".
N. New columns in a refreshed extract	"New columns since the last check: CURR_STATUS, DTI. Columns!C3 needs a Yes again."	Check them, set C3 to Yes.

A: two splits

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!H: only one column can split the pockets; ASSET_CLASS, REV_DEBT are all set to Yes.

B: the only segment split

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Nothing is left to cut across: mark at least one column as a category (or term) on the Columns tab, with Cut by it set to Yes. Changed from what was suggested: ASSET_CLASS (Columns!D13, Cut by it? No).

C: average on a category

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

Open the workbook

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Columns!G8: "CHANNEL" isn't a number column, so it has no average to show.
Clear the cell, or change what it is.

D: edges that Excel made into one number, then **D2: an edge outside the range**, then commas kept as text (it ran, with 4 bands):

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

Open the workbook

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Columns!F7: the band edges for "FICO" read as the single number 620,680,740.
Excel dropped the commas. Type them with semicolons: 620; 680; 740.

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Columns!F7: FICO runs from 496 to 921, so an edge at 950 would leave a band empty. Type edges inside that range, with semicolons: 620; 680; 740.

Check

Settings used, tie-outs, and what was left out.

Extract	Consumer book Q3.csv
Loans run	8,000
Record of this run	Consumer book Q3 - Origination Cube - what ran.yaml, beside this workbook: every setting the run used, kept for the file. It is replaced by the next Run.
Tie-out checks	132 of 132 agree: every grid adds up to the book
Left out of Outcome, share of loans	1 x BAD_FLAG flag not 0 or 1
Left out of Outcome, share of booked dollars	1 x ORIG_BAL blank; 1 x BAD_FLAG flag not 0 or 1
Left out of GCO per booked dollar	1 x GCO_AMT not a number; 1 x ORIG_BAL blank
Left out of RANR per booked dollar	1 x ORIG_BAL blank
Band edges used: FICO	620; 680; 740 (4 bands)
Band edges used: ORIG_BAL	15,449; 26,324; 37,951; 49,152 (5 bands)
Band edges used: REV_DEBT	6,559; 9,837; 13,051; 17,187 (5 bands)
Loans needed for a 1.25x gap: Outcome, share of loans	about 1,756

E: workbook open. Simulated by locking the file (chattr +i), since Excel isn't available here. Run and Set up both refuse, and neither the memory file nor the record file changes:

Origination Cube

Extract:

Consumer book Q3 - Origination Cube.xlsx is open in Excel. Close it, then press Run again.

Origination Cube

Extract:

Consumer book Q3 - Origination Cube.xlsx is open in Excel. Close it, then press Set up again.

F: a column renamed after Set up

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Couldn't run: the extract has no column "CHANNEL" (used by dimension channel).
Its columns are: LOAN_NBR, FICO, SALES_CHANNEL, ORIG_BAL, BAD_FLAG,
GCO_AMT, RANR_AMT, ASSET_CLASS, REV_DEBT

G: fewer rows after Set up

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Consumer book Q3.csv has changed since Set up. If columns were added or renamed, press Set up first.
Ran on 6,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 620 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 620 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 620 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,562 to under 9,791 / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

H: the workbook copied to another folder

Origination Cube

Extract:

Consumer book Q3.csv has changed since Set up. If columns were added or renamed, press Set up first.
Ran on 3,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 620 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: REV_DEBT 17,017 and over / CHANNEL Broker.
Worst for GCO per booked dollar: REV_DEBT 17,017 and over / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

I: split by GCO, and by the loan number

Origination Cube

Extract:

Ran on 8,000 loans from Consumer book Q3.csv; 677 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 620 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 620 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 620 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Split

Every pocket split in two at its own median GCO_AMT: the high half against the low half, holding the band and segment fixed. Above 1.00x means the high half does worse (for RANR, earns more). GCO_AMT isn't cut into bands of its own while it splits.

FICO x CHANNEL: Outcome, share of loans

The high-GCO_AMT half was worse in 1 of 14 pockets. Holding the pocket fixed, the high half loses 93.83x what it would at the low half's rate; chance it's luck under 0.0001.

High half vs low	Branch	Broker	Online
under 620		14.67x	
620 to under 680			
680 to under 740			
740 and over			
(blank)			
(marked missing)			

Chance it's luck	Branch	Broker	Online
under 620	under 0.0001	under 0.0001	under 0.0001
620 to under 680	under 0.0001	under 0.0001	under 0.0001
680 to under 740	under 0.0001	under 0.0001	under 0.0001
740 and over	under 0.0001	under 0.0001	under 0.0001
(blank)			
(marked missing)		under 0.0001	under 0.0001

FICO x CHANNEL: Outcome, share of booked dollars

The high-GCO_AMT half was worse in 1 of 14 pockets. Holding the pocket fixed, the high half loses 373.46x what it would at the low half's rate; chance it's luck under 0.0001.

High half vs low	Branch	Broker	Online
under 620		471.46x	

Chance it's luck	Branch	Broker	Online
under 620	under 0.0001	under 0.0001	under 0.0001

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

[Open the workbook](#)

Ran on 8,000 loans from Consumer book Q3.csv; 677 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 620 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 620 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 620 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

J: your own dollar materiality

Origination Cube

Extract: /tmp/claude-0/-home-user-SATC/b4305bfc-8769-5da1-bd67-

Browse...

1. Set up from this extract
2. Run the cube
- Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 620 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 620 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 620 / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Check

Settings used, tie-outs, and what was left out.

Extract	Consumer book Q3.csv
Loans run	8,000
Record of this run	Consumer book Q3 - Origination Cube - what ran.yaml, beside this workbook: every setting the run used, kept for the file. It is replaced by the next Run.
Tie-out checks	132 of 132 agree: every grid adds up to the book
Left out of Outcome, share of loans	1 x BAD_FLAG flag not 0 or 1
Left out of Outcome, share of booked dollars	1 x ORIG_BAL blank; 1 x BAD_FLAG flag not 0 or 1
Left out of GCO per booked dollar	1 x GCO_AMT not a number; 1 x ORIG_BAL blank
Left out of RANR per booked dollar	1 x ORIG_BAL blank
Band edges used: FICO	620; 680; 740 (4 bands)
Band edges used: ORIG_BAL	15,449; 26,324; 37,951; 49,152 (5 bands)
Band edges used: REV_DEBT	6,559; 9,837; 13,051; 17,187 (5 bands)
Loans needed for a 1.25x gap: Outcome, share of loans	about 1,756
Loans needed for a 1.25x gap: Outcome, share of booked dollars	about 2,203
Loans needed for a 1.25x gap: GCO per booked dollar	about 2,359
Loans needed for a 1.25x gap: RANR per booked dollar	about 411
Materiality line: Outcome, share of booked dollars	100,000 ORIG_BAL dollars
Materiality line: GCO per booked dollar	100,000 GCO_AMT dollars
Materiality line: RANR per booked dollar	100,000 RANR_AMT dollars
How old a loan must be to count	Every loan
Fewest loans in a pocket before it is tested	30 loans
Fewest loans with a loss before a loss rate is tested	10 losses
Smallest excess loss worth reporting	100000
What a pocket is judged against	The rest of its band
How much worse than its comparison a pocket must be	1.25 times
How much better than its comparison a pocket must be	0.8 times
How sure a difference must be before it counts	95% sure
How often a real gap should be caught	80% of the time
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg)
Warning	outcome_loans counts loans, not dollars, so the dollar materiality line is not applied to it

K: the workbook picked as the extract

Origination Cube

Extract:

Set up Consumer book Q3 - Origination Cube - Origination Cube.xlsx from
Consumer book Q3 - Origination Cube.xlsx: 11 loans, 4 columns.
11 things to look at first, on the Columns tab.

Next: fill in the shaded cells on Control and Columns, save, close, and press Run.

L: 95 for "how sure", and 0.9 for "how much worse"

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Control!D19: "How much worse than its comparison a pocket must be" needs a multiple above 1, such as 1.4, from 1.01 to 100; got 0.9.
- Control!D22: "How sure a difference must be before it counts" needs a share, such as 0.95 for 95%, from 0.5 to 0.999; got 95. For 95%, type 0.95.

N: new columns in a refreshed extract

Origination Cube

Extract: /tmp/claude-0/-home-user-SATC/b4305bfc-8769-5da1-bd67-

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv:

8,000 loans, 11 columns.

New columns since the last check: CURR_STATUS, DTI. Columns!C3 needs a Yes again.

2 things to look at first, on the Columns tab.

Next: fill in the shaded cells on Columns, save, close, and press Run.

Columns

For any meaning that's wrong, and set Cut by it to No for anything you don't want in the grid. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? ☐ New since the last check: CURR_STATUS, DTI. Check them, then set C3 to Yes again.

Column	What it is	Cut by it? Yes means (outcomes only)	Band edges (use ; between)	Show per pocket?	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)						Remembered: you confirmed this as loan or application number on 2026-09-25	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes	620; 680; 740				Remembered: you confirmed this as a FICO score on 2026-09-25	0%	-9999, 713, 716
CHANNEL	Category	Yes					Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	Online, Broker, Branch
ORIG_BAL	Booked amount	Yes					Remembered: you confirmed this as booked or loan amount, which weights the reporting-level rates on 2026-09-25	0%	7655.75, 27999.47, 39684.43
BAD_FLAG	Outcome (yes/no)					1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	Remembered: you confirmed this as the yes/no outcome: charged off, ever delinquent, decided by the system on 2026-09-25	0%	0, 2, 1
GCO_AMT	GCO dollars					1 value isn't a number among 7.999 that are. They're left out of any rate using this column and counted.	Remembered: you confirmed this as gross charge-off dollars on 2026-09-25	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars						Remembered: you confirmed this as RANR dollars on 2026-09-25	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes					Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	1, 2, 4
REV_DEBT	Amount or number	Yes					Remembered: you confirmed this as a number to cut into bands on 2026-09-25	0%	15868.86, 12884.7, 8920.97
CURR_STATUS	Not used					New since the last check.	one value only (Current)	0%	Current
DTI	Not used					New since the last check.	one value only (0.3)	0%	0.3

How this was walked

On a Linux test machine:

- **The window:** Origination Cube.pyw's window on a virtual display, its buttons pressed by driver/drive.py (through driver/win.sh), each state photographed.
- **Excel's part:** done cell by cell with openpyxl (driver/edit.py), standing in for picking from the dropdowns and typing.
- **The workbook:** each state printed through LibreOffice (driver/render.sh) and photographed page by page. driver/shots.py crops and rings every picture here.
- **Wrong turns:** each in a folder of its own, from a copy of the answered workbook (driver/wrong.sh).
- **The build:** frozen at commit 00c0bdd with git archive.

- **Memory and preferences:** a scratch CUBE_MEMORY and HOME.
- **The no-effect check** behind step 18's warning: driver/noplant.py.

The order walked: steps 1 to 17, then 18, 19, 20 and 21 on the same workbook, then 22 and 23, then the wrong turns.